JAMES ROBERT PATTERSON

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PROFESSIONAL SUMMARY

Research Engineer with 10 years of experience designing and implementing machine learning systems and distributed computing solutions. Specialised in optimising large-scale data pipelines and developing novel algorithms for real-time processing. Passionate about solving complex technical problems and mentoring junior engineers through hands-on collaboration. Currently based in Leeds and seeking opportunities to lead research initiatives in emerging technologies.

WORK EXPERIENCE

Senior Research Engineer

Nexus Analytics Ltd

nexus-analytics.io

Leeds, United Kingdom

March 2021 - Present

Led the design and implementation of a distributed graph processing framework that reduced query latency by 65% across 500+ nodes. Mentored a team of 5 junior engineers in best practices for scalable system design and code review standards. Published 2 peer-reviewed papers on efficient indexing strategies for time-series databases in major conferences. Architected and deployed a real-time anomaly detection pipeline processing 2 million events per second with sub-second latency requirements.

Research Engineer

Catalyst Technologies

catalyst-tech.co

Manchester, United Kingdom

June 2017 - February 2021

Developed core algorithms for natural language processing that improved model accuracy from 78% to 94% through novel feature engineering and ensemble methods. Designed microservices architecture handling 100,000 concurrent users with 99.99% uptime SLA. Optimised GPU utilisation across our ML training clusters reducing computation costs by 40% annually. Collaborated with product teams to translate research prototypes into production systems deployed to 2 million+ users.

Research Engineer

Innovate Systems Group

innovate-systems.net

Bradford, United Kingdom

September 2014 - May 2017

Built automated testing frameworks that reduced bug detection time from weeks to hours during the development cycle. Researched and implemented streaming algorithms for cardinality estimation in high-velocity data streams. Contributed to open source projects and established technical documentation standards adopted across the engineering team. Developed proof-of-concept systems for 3 client projects involving computer vision and signal processing applications.

EDUCATION

Doctor of Philosophy in Computer Science

University of Leeds

Leeds, United Kingdom

2014

Bachelor of Science in Computer Science with First Class Honours

University of Manchester

Manchester, United Kingdom

2011

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, Go, Scala

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, deep neural networks

Distributed Systems: Apache Spark, Kafka, Flink, Hadoop, Kubernetes

Databases: PostgreSQL, MongoDB, Cassandra, Redis, ClickHouse

Cloud Platforms: AWS, Google Cloud Platform, Azure

DevOps and Tools: Docker, Jenkins, Git, Linux, Apache Airflow

Data Processing: SQL, Pandas, NumPy, data warehousing, ETL pipelines

Specialisations: Scalable systems architecture, algorithm optimisation, performance tuning